DA346: Deep Reinforcement Learning

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Course Description

This course introduces the fundamental concepts and cutting-edge developments in Deep Reinforcement Learning (DRL). Students will begin with the classical theory of Search problems, Markov Decision Processes and value-based learning, progressing toward modern DRL techniques such as Deep Q-Networks (DQN), Policy Gradients, Actor-Critic methods, and Proximal Policy Optimization (PPO). Topics will include exploration vs exploitation, function approximation, policy search, and recent advances such as Decision Transformers, Offline RL, and Inverse Reinforcement Learning. The course blends rigorous mathematical understanding with practical implementation using Python, PyTorch and OpenAI Gym.

Prerequisite(s):

Probability, linear algebra, calculus, and prior coursework in machine learning. Python proficiency and basic deep learning (e.g., PyTorch) is required.

Credit Hours: 4

Textbooks and References

- **Deep Reinforcement Learning**, by Aske Plaat — available online at https://deep-reinforcement-learning.net/
- **Foundations of Deep Reinforcement Learning** by Laura Graesser and Wah Loon Keng.
- **Reinforcement Learning: An Introduction** (2nd Edition) by Sutton and Barto.
- **Deep Reinforcement Learning Hands-On** by Maxim Lapan.
- **Artificial Intelligence: A Modern Approach** (4th Edition) by Stuart Russell and Peter Norvig.

Course Objectives

1. Understand the theoretical foundations of reinforcement learning.
2. Implement classical RL algorithms (e.g., value iteration, Q-learning).
3. Explore the transition from tabular to deep RL.
4. Apply policy gradient methods, actor-critic models, and PPO.
5. Analyze and apply DRL to environments like games and robotics.
6. Evaluate and critique current research in deep RL.

Course Outcomes

1. Understand search problems and solve them via planning.
2. Formulate and solve problems using MDPs.
3. Build scalable RL systems with function approximators.
4. Compare model-free and model-based approaches.
5. Design, implement, and tune DRL agents.
6. Reproduce and extend current RL research methods.

Grade Distribution:

- **Quizzes and Assignments:** 30%
- **Midterm Exam:** 20%
- **Final Exam:** 20%
- **Final Project (Research or Implementation):** 30%

Weekly Course Outline

Week	Topics
1	Introduction to AI Planning: Uninformed and Informed Search
2	Adversarial Search and Constraint Satisfaction Problems (CSPs)
3	Introduction to MDPs, Bellman Equations and Reinforcement Learning
4	Dynamic Programming: Value and Policy Iteration
5	Monte Carlo and TD Learning (SARSA, Q-Learning)
6	Exploration-Exploitation, Epsilon-Greedy, UCB
7	Deep Q-Networks: DQN, Target Networks, Experience Replay
8	Policy Gradient Methods, REINFORCE
9	Actor-Critic Methods and A2C/A3C
10	Proximal Policy Optimization (PPO), TRPO
11	Applications: Atari, Robotics, Mujoco
12	Advanced Topics: Meta-RL, Multi-Agent RL
13	Imitation Learning and Inverse RL
14	Offline RL, Decision Transformers
15	Project Presentations and Course Wrap-up